

Longitudinal Analysis of HPC Utilization at a Research University: Trends, Insights, and Data-Driven Interventions for Optimized Resource Allocation

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Abstract

The growing adoption of high-performance computing (HPC) resources across diverse research disciplines has created an urgent need for data-informed approaches to resource allocation and support deployment. This paper presents a longitudinal analysis of utilization patterns on HOPPER, the primary HPC cluster at George Mason University (GMU), spanning three academic periods: AY 2023–2024, the full calendar year 2025, and the Fall 2025–Spring 2026 academic semester window. Drawing on SLURM job scheduler records comprising over 7.4 million job submissions, we characterize usage trends across compute resource types, institutional colleges, academic departments, user cohorts, and instructional accounts. Our analysis reveals three structural shifts: (1) a rapid acceleration of GPU workloads, with daily GPU job counts quadrupling between early 2025 and Spring 2026 and GPU-hour consumption becoming highly concentrated among a small number of power users; (2) a near-tripling of College of Science (COS) utilization accompanied by the emergence of GPU adoption in natural science disciplines; and (3) the rise of the IST department as the single largest job submitter by volume, exhibiting a fundamentally different usage profile from other major consumers. We further identify bimodal CPU request distributions indicating systematic under-utilisation of shared-memory parallelism and recurring semester-correlated queue congestion events reaching average wait times of 4.3 hours.

Based on these findings we propose evidence-based interventions spanning personnel specialisation, queue policy design, infrastructure planning, and targeted outreach.

Keywords: high-performance computing; HPC utilization; resource allocation; GPU computing; Slurm; research computing; academic HPC

1 Introduction

High-performance computing has become an essential instrument across the full breadth of academic research, from the traditionally computation-intensive fields of physics and engineering to disciplines such as the life sciences, social science, and the humanities, where computational methods have only recently become mainstream [10, 15]. Institutional HPC centres must therefore serve an increasingly heterogeneous user population while managing constrained hardware and personnel resources. The challenge is not simply providing sufficient compute capacity; it is deploying that capacity—and the human expertise needed to use it productively—in ways calibrated to actual user needs and behaviour.

Most analyses of HPC usage address this challenge from the perspective of aggregate throughput: total CPU-hours consumed, overall system utilisation rates, and job success ratios. These measures are valuable for justifying institutional investment but provide limited guidance for operational decision-making [1, 16]. A growing body of work has demonstrated that disaggregated analysis—examining usage by discipline, user role, job type, and temporal pattern—yields actionable insights invisible in aggregate statistics [4, 14]. Understanding that a particular department submits thousands of very short jobs while another submits dozens of multi-day parallel simulations has direct implications for queue configuration, support priority, and training content, even when the aggregate resource consumption of the two groups is similar.

This paper extends the framework introduced in our earlier work on optimising HPC resource deployment at GMU [5] through a systematic longitudinal analysis of HOPPER cluster utilisation spanning from the 2023–2024 academic year through Spring 2026. Three findings motivate the core contributions. First, GPU workloads have grown dramatically and now exhibit levels of concentration—a single user consuming more GPU-hours than the next six combined over a nine-month window—that raise questions about equitable access in a shared resource environment. Second, the College of Science, historically a secondary user of HOPPER primarily running CPU-based simulations, has nearly tripled its utilisation and is beginning to adopt GPU workflows in atmospheric modelling, structural biology, and chemistry. Third, the IST department has become the largest single job submitter in the university by count while simultaneously having the lowest average per-job resource consumption, a mismatch that challenges conventional assumptions about who the “heavy users” are and what form effective support should take.

The paper proceeds as follows. Section 2 reviews related work. Section 3 describes the HOPPER system and data sources. Section 4 presents the analytical methodology. Section 5 reports the empirical findings. Section 6 interprets these findings. Section 7 translates the analysis into specific, evidence-based interventions. Section 10 concludes.

2 Background and Related Work

2.1 HPC Utilisation Analysis

The systematic analysis of HPC usage statistics has a well-established tradition in the research computing community. Early work focused primarily on demonstrating the productivity impact of HPC investment: Smith [15] showed that computational researchers produce statistically significantly more research output than non-computational peers across disciplines. Furlani et al. [4] developed the XDMoD framework, which operationalises a broad range of utilisation metrics enabling systematic analysis at both national XSEDE scale and at individual institutional centres. Hammond [7]

extended this approach to I/O performance monitoring.

More recent work has examined utilisation patterns in relation to user support and training. Kilpatrick et al. [9] demonstrated that users who receive targeted consultation from research computing facilitators show measurable improvements in job efficiency. A complementary line of research has examined the relationship between system complexity and user error rates, finding that the barrier of command-line SLURM job submission is a major source of both inefficiency and support burden [12]. The deployment of web-based interfaces such as Open OnDemand has been shown to reduce this barrier and increase effective cluster utilisation among non-specialist researchers [8].

2.2 GPU Computing in Academic HPC

The rapid diffusion of GPU-accelerated workloads into academic HPC environments has introduced new challenges for resource management. Netti et al. [11] analysed GPU utilisation across a large European HPC facility and found that a small fraction of users—typically 10–15%—account for 60–80% of GPU-hours consumed, a concentration pattern consistent with what we observe at GMU. Chard et al. [3] examined the impact of GPU demand growth on queue dynamics, showing that wait-time inflation during peak periods is disproportionately driven by large-GPU-count jobs submitted by a small number of users.

2.3 Instructional HPC Use

The use of HPC clusters for formal coursework represents a qualitatively distinct usage pattern from research computing. Instructional jobs are characterised by high submission synchronicity, short individual job duration, and strong temporal correlation with the academic calendar [13]. The expansion of data science and machine learning curricula has accelerated the instructional demand for GPU resources specifically [17].

2.4 Resource Allocation Under CaRCC Frameworks

The Campaign for Research Computing Competencies (CaRCC) has articulated a structured framework for classifying research computing personnel activities into four facings: researcher-facing, system-facing, software/data-facing, and sponsor/stakeholder-facing [2]. Our earlier work [5] introduced the linkage between utilisation analysis and CaRCC-structured personnel deployment; the present analysis provides the longitudinal data needed to assess how the composition of support needs has changed over time.

3 System Description and Data Sources

3.1 The Hopper Cluster

HOPPER is the primary HPC cluster operated by the GMU Office of Research Computing (ORC). At the time of this analysis, HOPPER comprises approximately 11,840 CPU cores, a VAST flash-based scratch storage system, and a heterogeneous GPU partition including NVIDIA A100 (40 GB and 80 GB), A40, H100, and B200 class GPUs accessed via MIG and full-device configurations. Users access HOPPER via SSH connections to login nodes or through the ORC Open OnDemand (OOD) web interface [8]. Job scheduling is managed by the SLURM Workload Manager [18].

3.2 Data Sources and Coverage

The primary data source is the SLURM job accounting database, from which records were extracted covering three non-overlapping periods:

- **AY 2023–2024:** July 2023 through June 2024 (12 months, baseline period).
- **Full Year 2025:** January 2025 through December 2025 (12 months, retrospective period).
- **Fall 2025–Spring 2026:** August 2025 through April 2026 (9 months, current period).

For each completed job the extracted record includes: user identifier (anonymised), account identifier, submit time, start time, end time, requested CPUs (NCPUS), requested GPUs (NGPUS), requested nodes (NNODES), requested memory (GB), consumed CPU-hours (CPUHRS), consumed GPU-hours (GPUHRS), and job exit state. Instructional accounts are identifiable by a naming convention encoding the course prefix, number, and semester code (e.g., `cs678f125` for CS 678, Fall 2025).

4 Methodology

4.1 Metrics

We report the following utilisation metrics, computed per job and aggregated over monthly, semester, or annual time windows:

- **Total jobs:** count of completed records, disaggregated by CPU vs. GPU (jobs with $NGPUS > 0$).
- **CPU-hours (CPUHRS):** product of allocated CPUs and elapsed wall time.
- **GPU-hours (GPUHRS):** product of allocated GPUs and elapsed wall time.
- **Average memory requested:** mean allocation in GB per job.
- **Average wall time and average wait time:** mean elapsed and queueing durations in hours.
- **Active unique users:** distinct user identifiers submitting at least one job in a given time window.

4.2 Classification and Longitudinal Comparison

Jobs are classified as *CPU jobs* if $NGPUS = 0$ and *GPU jobs* if $NGPUS > 0$. Institutional affiliations are classified at college and department levels. Instructional accounts are treated as a distinct stratum. Longitudinal comparisons normalise to a per-month basis where windows of different length are compared.

5 Results

5.1 Overall Utilisation Trends

Total job volume increased substantially across the three analysis periods. The Full Year 2025 retrospective recorded **2,283,321** total job submissions (CPU: 2,159,248; GPU: 124,073). The Fall 2025–Spring 2026 nine-month window recorded **2,905,604** total submissions (CPU: 2,777,817; GPU: 127,787). That a nine-month period exceeds the full 2025 yearly total confirms a strong upward volume trend. Figure 1 shows the aggregate statistics across all three periods.

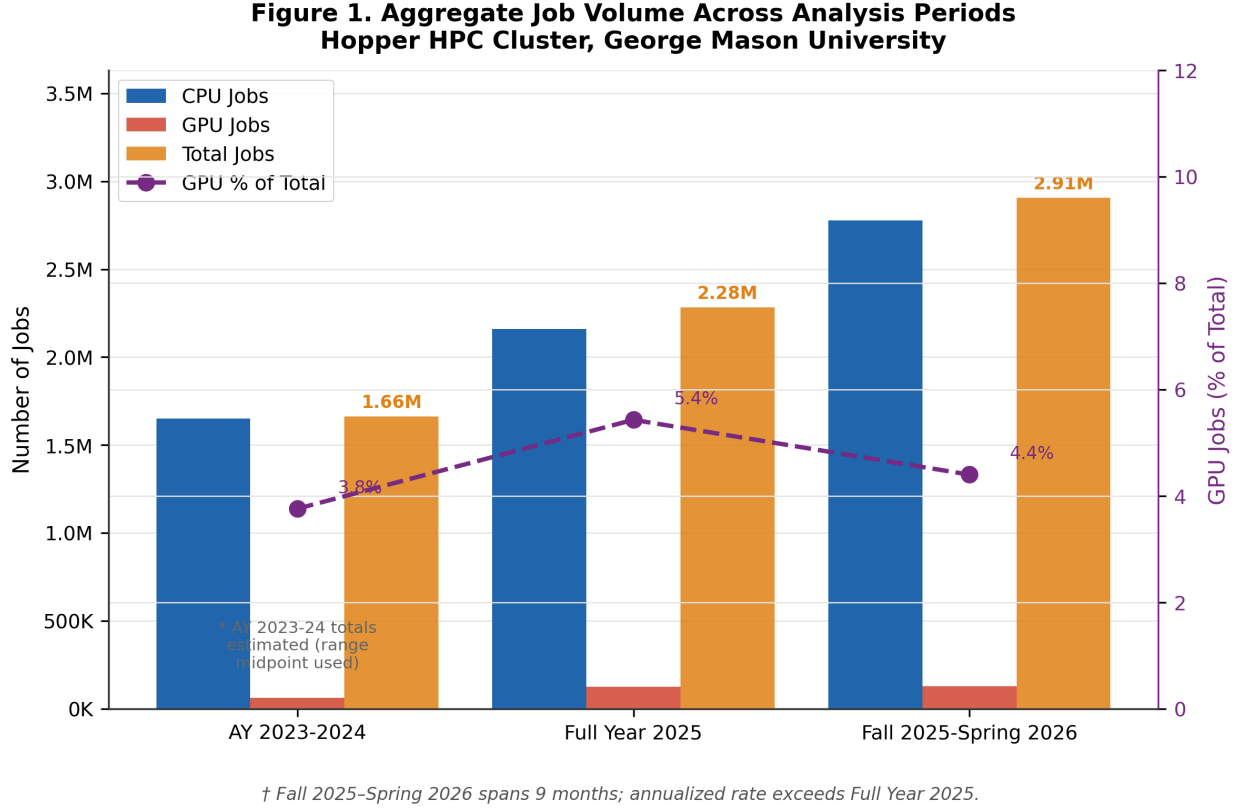


Figure 1: Aggregate job volume across all three analysis periods. CPU jobs, GPU jobs, and totals are shown as grouped bars (left axis); the GPU percentage of total submissions is plotted as a connected line series (right axis, purple). AY 2023–2024 totals are estimated from the reported range midpoint. The Fall 2025–Spring 2026 bar spans only nine months; its annualised rate exceeds the Full Year 2025 total.

Table 1 summarises the aggregate statistics.

Table 1: Aggregate utilisation statistics across analysis periods.

Period	Months	CPU Jobs	GPU Jobs	Total Jobs	GPU %
AY 2023–2024	12	~1.65M*	~62.5K*	~1.66M*	~3.8%
Full Year 2025	12	2,159,248	124,073	2,283,321	5.4%
Fall 2025–Spr. 2026	9	2,777,817	127,787	2,905,604	4.4%

* Estimated from range midpoint; exact aggregate not extracted in original analysis.

Monthly job submission patterns reveal pronounced seasonality. In Full Year 2025, September (367,874 CPU jobs), October (323,931), and February (312,983) were the highest-volume months. July (58,043 CPU jobs) was the lowest, confirming a summer trough consistent with patterns at comparable academic facilities [13]. Average daily unique active users peaked at 617 on Tuesdays in the 2025 data. Weekend averages remained substantial (506–514 per day in 2025; 320–323 per day in Fall 2025–Spring 2026), indicating a large self-directed user population.

5.2 College-Level Utilisation

Figure 2 presents the distribution of job submissions by institutional college in the Fall 2025–Spring 2026 period, and Table 2 provides the numerical breakdown.

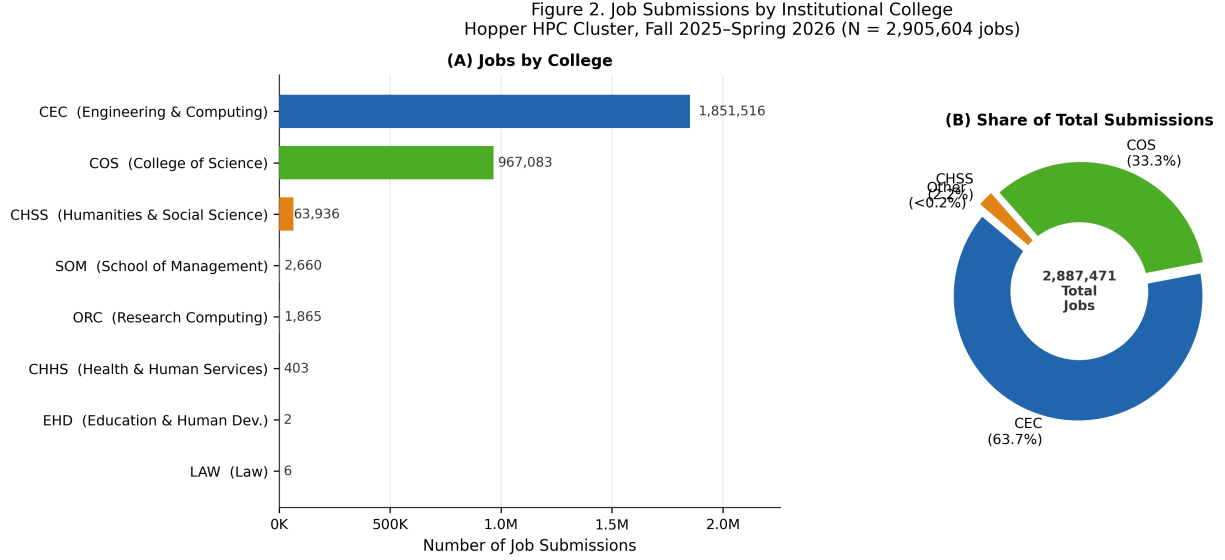


Figure 2: Job submissions by institutional college, Fall 2025–Spring 2026 (N = 2,905,604). Panel (A): horizontal bar chart of absolute job counts. Panel (B): donut chart showing the share of total submissions; small colleges (SOM, ORC, CHHS, EHD, LAW) are aggregated as “Other” (<0.2%).

Table 2: Job submissions by institutional college, Fall 2025–Spring 2026.

College	Jobs	Share (%)
CEC (Engineering & Computing)	1,851,516	63.7
COS (College of Science)	967,083	33.3
CHSS (Humanities & Social Science)	63,936	2.2
SOM (School of Management)	2,660	0.1
Other (ORC, CHHS, EHD, LAW)	<2,300	<0.1

The COS figure represents a qualitative shift: in Full Year 2025, COS submitted 342,096 CPU jobs over 12 months. The subsequent nine-month window records 967,083 jobs—a 2.8-fold increase in annualised rate—indicating that COS has undergone a phase transition from a secondary to a co-primary user of HOPPER resources.

5.3 Department-Level Analysis: CEC

Figure 3 presents department-level resource profiles. Within CEC, Table 3 summarises CPU job statistics.

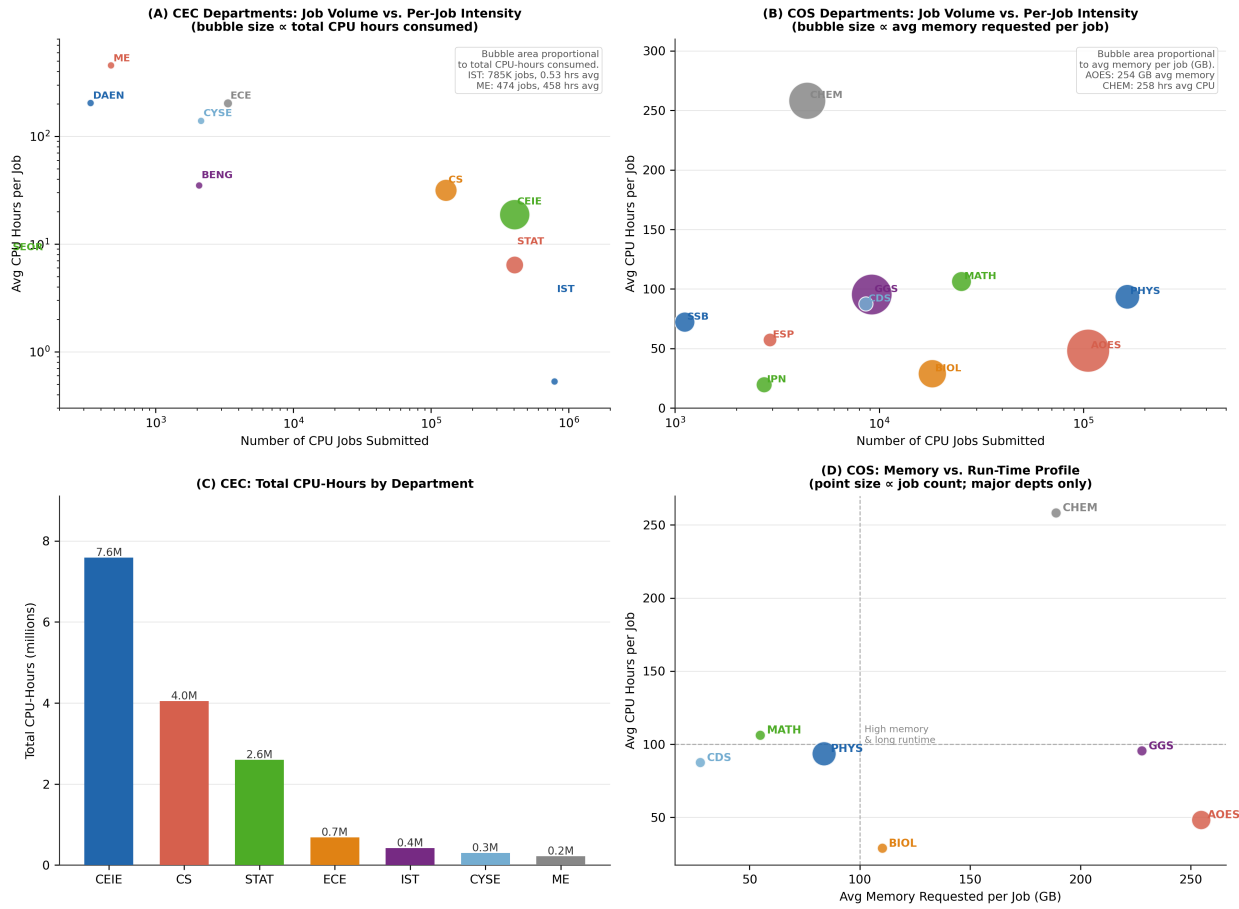
Figure 3. Departmental Resource-Use Profiles — CEC and COS
Hopper HPC Cluster, Fall 2025–Spring 2026

Figure 3: Departmental resource-use profiles for CEC (Panels A, C) and COS (Panels B, D). Bubble charts (A, B) plot job count vs. average CPU-hours per job on logarithmic axes; bubble area is proportional to total CPU-hours (A) or average memory per job (B). Bar charts (C, D) compare total CPU-hours across departments and, for COS, plot average memory against average run time.

Table 3: CEC CPU job statistics by department, Fall 2025–Spring 2026.

Department	CPU Jobs	Total CPU Hrs	Avg CPU Hrs/Job	Avg Mem (GB)
IST (Info. Sciences & Tech.)	785,603	417,913	0.53	4.10
STAT (Statistics)	404,830	2,598,905	6.42	13.30
CEIE (Civil & Env. Eng.)	403,909	7,592,789	18.80	35.27
CS (Computer Science)	128,234	4,044,583	31.54	34.61
CYSE (Cybersecurity Eng.)	2,138	298,126	139.44	221.67
ECE (Electrical & Computer Eng.)	3,339	679,437	203.49	32.56
DAEN (Data Analytics Eng.)	337	68,951	204.60	48.94
ME (Mechanical Engineering)	474	217,121	458.06	143.81

The most striking feature of Table 3 is the contrast between IST and all other CEC departments. IST is the single largest job submitter in the entire university (785,603 CPU job records), yet each IST

job runs for an average of only 0.53 hours and requests 4.1 GB of memory. This profile is consistent with high-throughput automated workflows or classroom exercises rather than long-running research simulations. At the opposite extreme, Mechanical Engineering averages 458 CPU-hours per job—a factor of ~ 865 higher than IST—and Data Analytics Engineering and ECE similarly display high per-job intensity at low submission volume.

GPU usage within CEC is dominated by Computer Science (57,388 GPU jobs; 424,340 GPU-hours) and IST (33,244 GPU jobs; 220,621 GPU-hours) as shown in Table 4.

Table 4: CEC GPU job statistics by department, Fall 2025–Spring 2026.

Department	GPU Jobs	Total GPU Hrs	Avg GPU Hrs/Job	Avg GPU Mem (GB)
CS	57,388	424,340	7.39	79.20
IST	33,244	220,621	6.64	110.84
ECE	16,750	74,279	4.43	41.85
SEOR	1,077	25,688	23.85	86.54
CYSE	3,437	7,003	2.04	90.16

IST’s average GPU memory request of 110.84 GB per job is particularly notable, as it implies that a significant fraction of IST GPU jobs request A100 80 GB, H100, or B200 instances.

5.4 Department-Level Analysis: COS

Table 5 presents CPU job statistics for COS departments.

Table 5: COS CPU job statistics by department, Fall 2025–Spring 2026.

Department	CPU Jobs	Total CPU Hrs	Avg CPU Hrs/Job	Avg Mem (GB)
PHYS (Physics & Astronomy)	164,088	15,371,948	93.68	83.78
AOES (Atmospheric, Ocean & Earth Sci.)	105,579	5,106,175	48.36	254.60
MATH (Mathematical Sciences)	25,265	2,686,843	106.35	54.78
BIOL (Biology)	18,187	525,899	28.92	110.07
GGG (Geography & Geo-Information Sci.)	9,192	877,814	95.50	227.83
CDS (Computational & Data Science)	8,587	751,942	87.57	27.58
CHEM (Chemistry & Biochemistry)	4,439	1,146,580	258.30	188.83

The COS profile differs from CEC in two important respects. First, per-job resource intensity is substantially higher: Chemistry averages 258.30 CPU-hours per job and AOES averages 254.60 GB of memory per job. Second, COS GPU adoption—essentially zero in Full Year 2025 (2 total GPU jobs)—reached approximately 8,300 GPU jobs by Fall 2025–Spring 2026 (Table 6).

Table 6: COS GPU job statistics by department, Fall 2025–Spring 2026. Note: COS submitted only ~2 GPU jobs in the entirety of Full Year 2025.

Department	GPU Jobs	GPU Hrs	Avg GPU Hrs/Job	Avg GPU Mem (GB)
AOES	3,654	35,157	9.62	133.88
SSB (School of Systems Biology)	2,864	62,570	21.85	27.46
CDS	558	8,782	15.74	38.10
CHEM	305	15,786	51.76	68.43
GGs	459	1,597	3.48	75.97
MATH	240	4,260	17.75	160.34
PHYS	164	671	4.09	48.72

5.5 GPU Workload Analysis

Figure 4 presents four panels of GPU workload analysis: departmental distribution, COS GPU emergence, top-user concentration, and the GPU-hour Lorenz curve.

Figure 4. GPU Workload Analysis: Departmental Distribution, COS Emergence, User Concentration, and Lorenz Curve — Hopper HPC Cluster

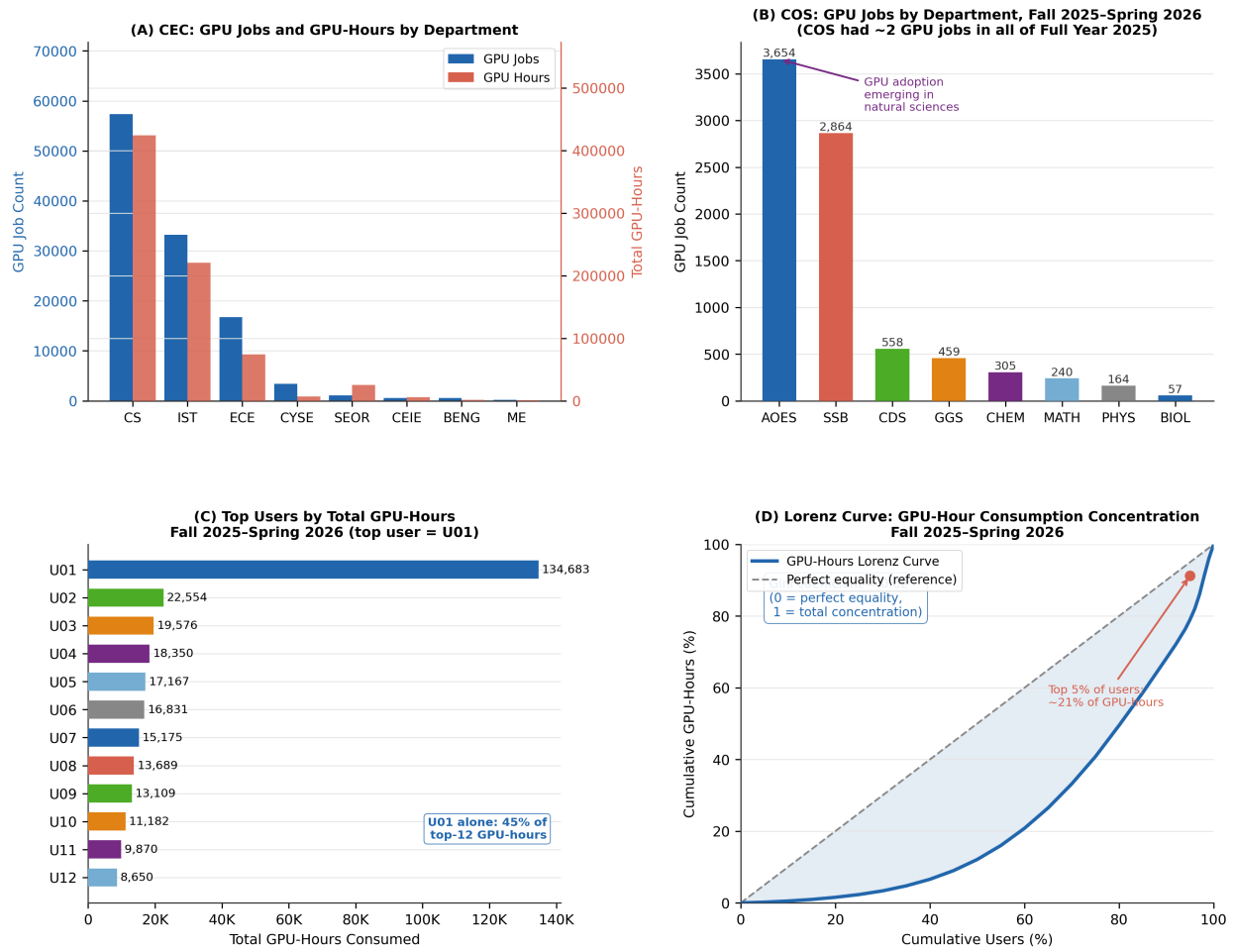


Figure 4: GPU workload analysis, Fall 2025–Spring 2026. **(A)** CEC GPU jobs and GPU-hours by department (dual y-axis). **(B)** COS GPU jobs by department; the subtitle notes that COS submitted only ~2 GPU jobs in all of Full Year 2025. **(C)** Top 12 users by total GPU-hours consumed (anonymised as U01–U12); U01 alone accounts for 45 % of the top-12 total. **(D)** Lorenz curve of GPU-hour consumption; the shaded area between the curve and the equality diagonal reflects an approximate Gini coefficient of 0.62.

Table 7 presents the top GPU users by total GPU-hours.

Table 7: Top users by total GPU-hours, Fall 2025–Spring 2026. All user identifiers are anonymised.

User ID	Total GPU Hrs	GPU Jobs	Avg GPU Hrs/Job	Avg GPU Mem (GB)
U01	134,683	~2,513	~54	392.5
U02	22,554	~2,875	~8	64.8
U03	19,576	~271	~72	363.1
U04	18,350	5,233	3.5	39.9
U05	17,167	~241	~71	304.6
U06	16,831	~196	~86	429.1

The concentration of GPU-hours in the top user is extreme: U01 alone consumed more GPU-hours in the nine-month window than all COS departments combined. This pattern is consistent with findings at comparable European HPC facilities [11], where the top 10–15 % of users account for 60–80 % of GPU-hours.

5.6 Instructional Utilisation

Figure 5 presents four panels characterising instructional GPU use, and Table 8 lists the top accounts by job count.

Figure 7. Instructional GPU Utilization — Hopper HPC Cluster
Fall 2025–Spring 2026: Growing Course Demand and Hardware Contention

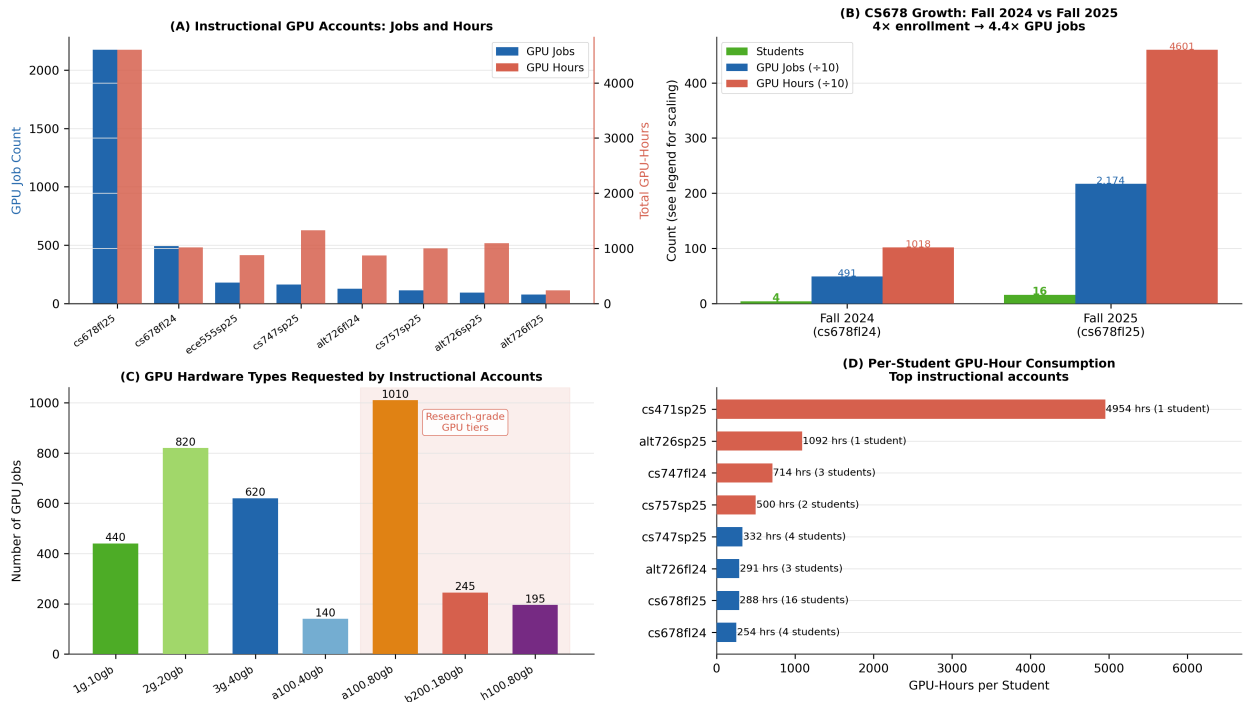


Figure 5: Instructional GPU utilisation, Fall 2025–Spring 2026. (A) Top instructional accounts by GPU jobs and GPU-hours. (B) Year-over-year growth for CS678: Fall 2024 vs. Fall 2025, showing 4× enrollment driving 4.4× GPU jobs. (C) GPU hardware types requested by course accounts; note requests for research-grade A100 80 GB, B200 180 GB, and H100 80 GB. (D) Per-student GPU-hour consumption for top accounts.

Table 8: Top instructional GPU accounts by job count, Fall 2025–Spring 2026. The final row provides the preceding year’s CS678 record for comparison.

Account	Students	GPU Jobs	GPU Hrs	Avg GPU Hrs/Job
cs678fl25	16	2,174	4,601	2.12
alt726sp25	1	94	1,092	11.63
cs471sp25	1	72	4,954	68.81
cs747sp25	4	162	1,330	8.21
cs757sp25	2	114	1,001	8.79
cs678fl24 [†]	4	491	1,018	2.07

[†] Prior-year comparison record.

5.7 CPU Request Distribution

Figure 6 reveals a pronounced bimodal structure in the distribution of CPU cores requested per job.

Figure 6. Bimodal CPU Request Distribution — Hopper HPC Cluster
Fall 2025–Spring 2026 (N = 2,777,817 CPU jobs)

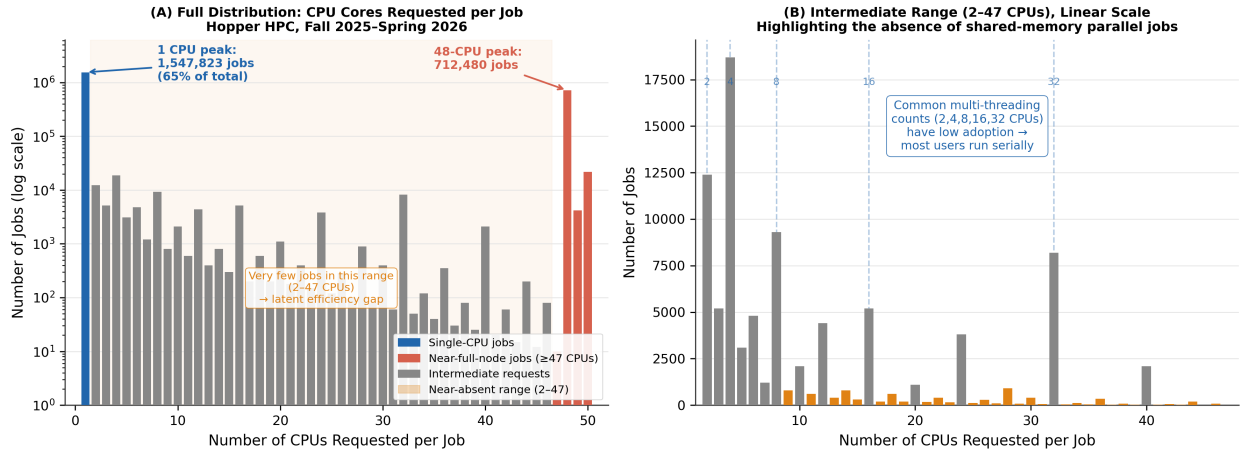


Figure 6: Distribution of CPU cores requested per job, Fall 2025–Spring 2026 ($N = 2,777,817$ CPU jobs). **(A)** Full distribution on a logarithmic y-axis. The dominant peak at 1 CPU (1,547,823 jobs; 65 % of total) and the secondary peak near 48 CPUs (712,480 jobs) are highlighted. **(B)** Zoomed view of the 2–47-CPU range on a linear scale, revealing the near-complete absence of intermediate multi-core requests.

The dominant peak at 1 CPU accounts for approximately 65 % of all CPU jobs. A secondary peak at 48 CPUs (approximately one full node on standard HOPPER configurations) contains a further 25 %. The intermediate range of 2–47 CPUs is nearly absent, indicating that most researchers either run serially or jump directly to full-node allocations, with very little adoption of intermediate-scale shared-memory parallelism.

5.8 Queue Health and Temporal Patterns

Figure 7 presents temporal utilisation patterns: monthly job volume for Full Year 2025 and monthly average wait and run times for Fall 2025–Spring 2026. Table 9 provides the numerical wait/run-time

data.

Figure 5. Temporal Utilization Patterns and Queue Health
Hopper HPC Cluster, George Mason University

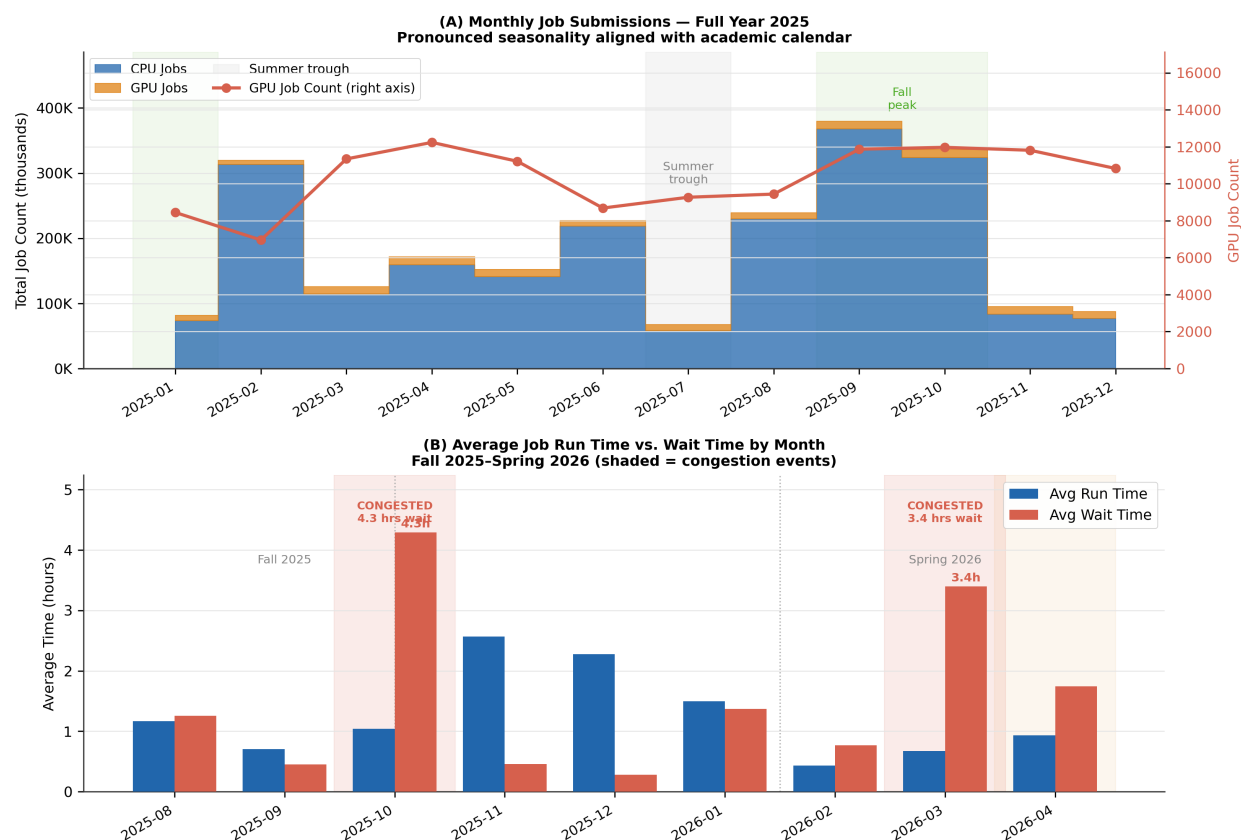


Figure 7: Temporal utilisation patterns. (A) Monthly CPU and GPU job submissions, Full Year 2025, with the summer trough and fall peak annotated. The GPU job count trend is overlaid on the right axis (red line). (B) Average job run time (blue) and wait time (red) by month, Fall 2025–Spring 2026. Months classified as “congested” (October 2025 and March 2026) are highlighted with a red background shade. Vertical dashed lines mark the Fall 2025/Spring 2026 semester boundary.

Table 9: Monthly average run time and wait time, Fall 2025–Spring 2026.

Month	Avg Run (hrs)	Avg Wait (hrs)	Status
2025-08	1.17	1.25	Normal
2025-09	0.71	0.45	Low
2025-10	1.04	4.29	Congested
2025-11	2.56	0.46	Normal
2025-12	2.28	0.28	Low
2026-01	1.50	1.37	Normal
2026-02	0.43	0.77	Normal
2026-03	0.68	3.40	Congested
2026-04	0.93	1.74	Elevated

Two months exhibit substantially elevated average wait times: October 2025 (4.29 hours) and

March 2026 (3.40 hours), corresponding to predictable points in the academic calendar: the first major research crunch of each semester.

Figure 8 presents daily active users and the CEC IST outlier profile.

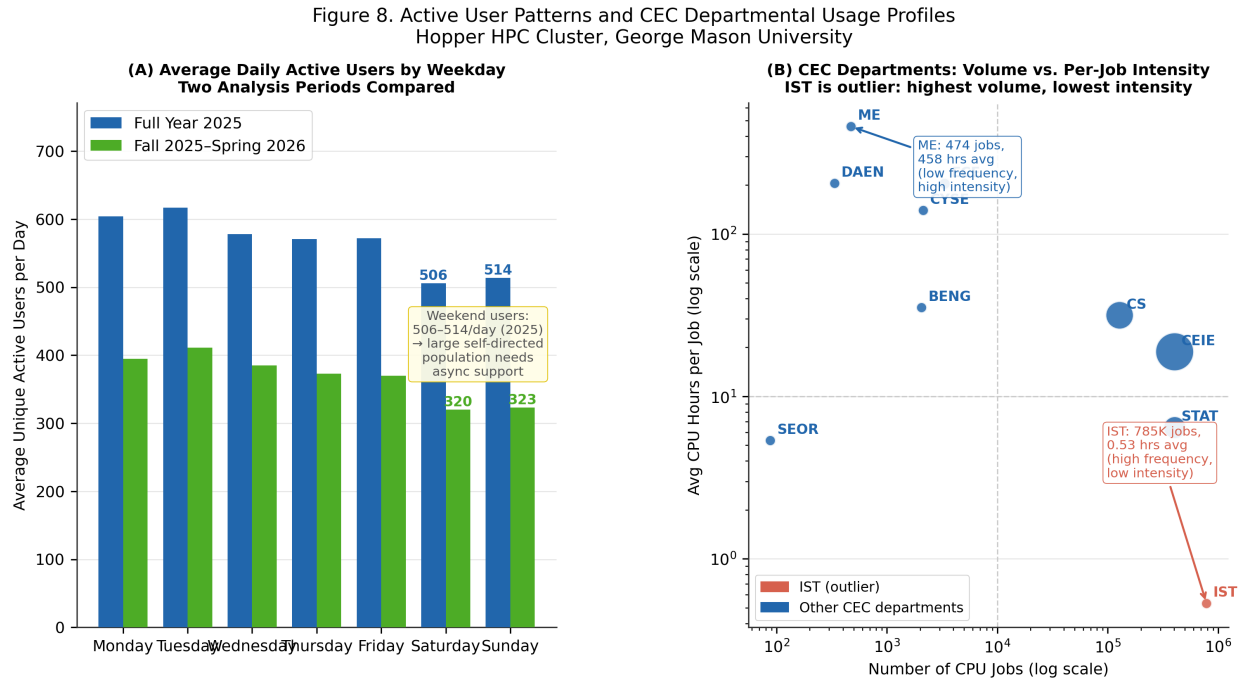


Figure 8: Active user patterns and CEC departmental usage profiles. **(A)** Average unique daily active users by day of week, comparing Full Year 2025 (blue) and Fall 2025–Spring 2026 (green). Weekend counts of 320–514 users per day confirm a large self-directed user population. **(B)** CEC departments plotted by CPU job count (x-axis) vs. average CPU-hours per job (y-axis), both on logarithmic scales. IST (highlighted in red) is a clear outlier: the highest volume department with the lowest per-job intensity. Bubble area is proportional to total CPU-hours consumed.

6 Discussion

6.1 A Structurally Transformed User Community

The data collectively indicate that the HOPPER user community has undergone a structural transformation, not merely quantitative growth. IST’s emergence as the largest job submitter—with a usage profile that differs categorically from all other major departments—is the clearest illustration of this diversity. IST’s workload appears driven by automated analysis pipelines, classroom exercises, or workflow frameworks generating thousands of short jobs rather than the intensive long-running computations traditionally associated with HPC. From a scheduling perspective, IST’s workload is not particularly resource-intensive; but thousands of short jobs submitted in bursts can inflate queue length and interfere with fair scheduling for users submitting fewer but larger jobs.

6.2 The GPU Transition: Acceleration, Concentration, and Spillover

GPU utilisation has entered a new phase characterised by three concurrent developments: overall demand growth, extreme user-level concentration, and cross-disciplinary spillover into natural science

fields. GPU daily job counts roughly quadrupled between early 2025 and Spring 2026. A single user consuming more GPU-hours than the next six combined constitutes a structural inequity that conventional first-come, first-served scheduling cannot address. The emergence of GPU workloads in COS—essentially zero in 2025, approximately 8,300 jobs by Spring 2026—represents the earliest phase of a GPU adoption curve in the natural sciences, with atmospheric modelling, protein structure prediction, and ML interatomic potentials as the leading applications.

6.3 Instructional GPU Demand and Hardware Contention

Students in CS678, CS471, and similar courses are requesting access to A100 80 GB, H100, and B200 hardware that is simultaneously the most demanded resource for research GPU users. The increase in CS678 enrolment from 4 students in Fall 2024 to 16 students in Fall 2025—a $4\times$ increase in a single year—suggests that instructional GPU demand is not simply growing but accelerating. Without a hardware partitioning strategy, the conflict between instructional and research GPU use will intensify predictably with each semester.

6.4 The Bimodal CPU Distribution as a Latent Efficiency Opportunity

The near-complete absence of jobs requesting 2–47 CPUs is unusual and warrants attention beyond its statistical interest. OpenMP, Python multiprocessing, and R parallel computing frameworks can provide $4\text{--}16\times$ speedups for many common workflows with minimal code modification. Targeted training in intermediate-scale parallelism for IST and STAT populations—the departments with the largest numbers of single-CPU job submitters—represents a high-leverage, low-cost intervention.

6.5 Semester-Synchronised Congestion

The October and March wait-time spikes (4.29 and 3.40 hours, respectively) are predictable consequences of semester-correlated research activity and are unlikely to resolve through organic scheduling behaviour. Proactive queue management—coordinated advance notice to the heaviest users before congestion events materialise—is a well-established intervention in operational HPC support [12].

7 Data-Driven Interventions

7.1 Personnel Specialisation

The scale and complexity of GPU workloads now justify a dedicated **GPU Specialist** within the ORC researcher-facing staff, requiring deep competency in CUDA, distributed GPU training frameworks (PyTorch DDP, JAX pmap, DeepSpeed), GPU memory profiling (DCGM, Nsight), and containerised GPU environments. The COS utilisation growth and GPU emergence requires a dedicated **COS Liaison** with competency in domain workflows: WRF/MOM6, GROMACS/NAMD/AMBER, AlphaFold, and geospatial computing. Given that IST generates more job records than any other department, an **IST/CS Instructional Coordinator** focused on semester-start onboarding, SLURM template development, and per-course quota planning would have disproportionate impact.

7.2 Queue and Scheduling Policy

We recommend a per-user GPU-hour soft cap on a rolling window (e.g., 10,000 GPU-hours per week), above which jobs receive lower scheduling priority rather than being blocked. Course accounts should

be restricted to lower-tier GPU partitions (1g.10gb, 2g.20gb, 3g.40gb via MIG) unless documented educational justification for high-memory devices is approved. Jobs requesting more than 512 CPUs, 72 hours of wall time, or 2 TB of memory should require a pre-submission consultation. Proactive semester-aligned outreach to the top 100 job submitters two weeks before each predicted congestion event (early October, mid-March) should be implemented as a standing operational procedure.

7.3 Infrastructure Planning

At current growth rates, GPU queue pressure will become a chronic operational constraint within one to two academic years. Hardware procurement should prioritise additional A100 80 GB or H100 80 GB nodes for research, with MIG partitioning for instructional access. The large memory demands in AOES (254 GB avg) and GGS (228 GB avg) suggest the high-memory node pool should be assessed for adequacy. Open OnDemand additions immediately justified by the data include GPU-enabled Jupyter environments and an interactive GPU session option for debugging.

7.4 Outreach and Training

A focused half-day workshop targeting IST and STAT users on Python multiprocessing, joblib, and array-job frameworks would address the bimodal CPU distribution at scale. COS GPU onboarding should begin now, targeting AOES (JAX for atmospheric modelling, neural-network potentials) and SSB (AlphaFold, genomic deep learning) while these communities are in early adoption.

8 Job Failure Rate Analysis

This section presents analysis A2: a systematic examination of job exit states, the wasted compute resources attributable to failure modes, and the relationship between job failure rates and queue congestion.

8.1 Exit-State Distribution by Department

Figure 9 presents four panels characterising job failure patterns across departments and time. Panel A shows the percentage breakdown of exit states for the nine highest-volume departments in Fall 2025–Spring 2026.

Figure A2. Job Failure Rate Analysis — Hopper HPC Cluster
Exit-State Distribution, Temporal Correlation with Congestion, Wasted Resources, and Over-Provisioning

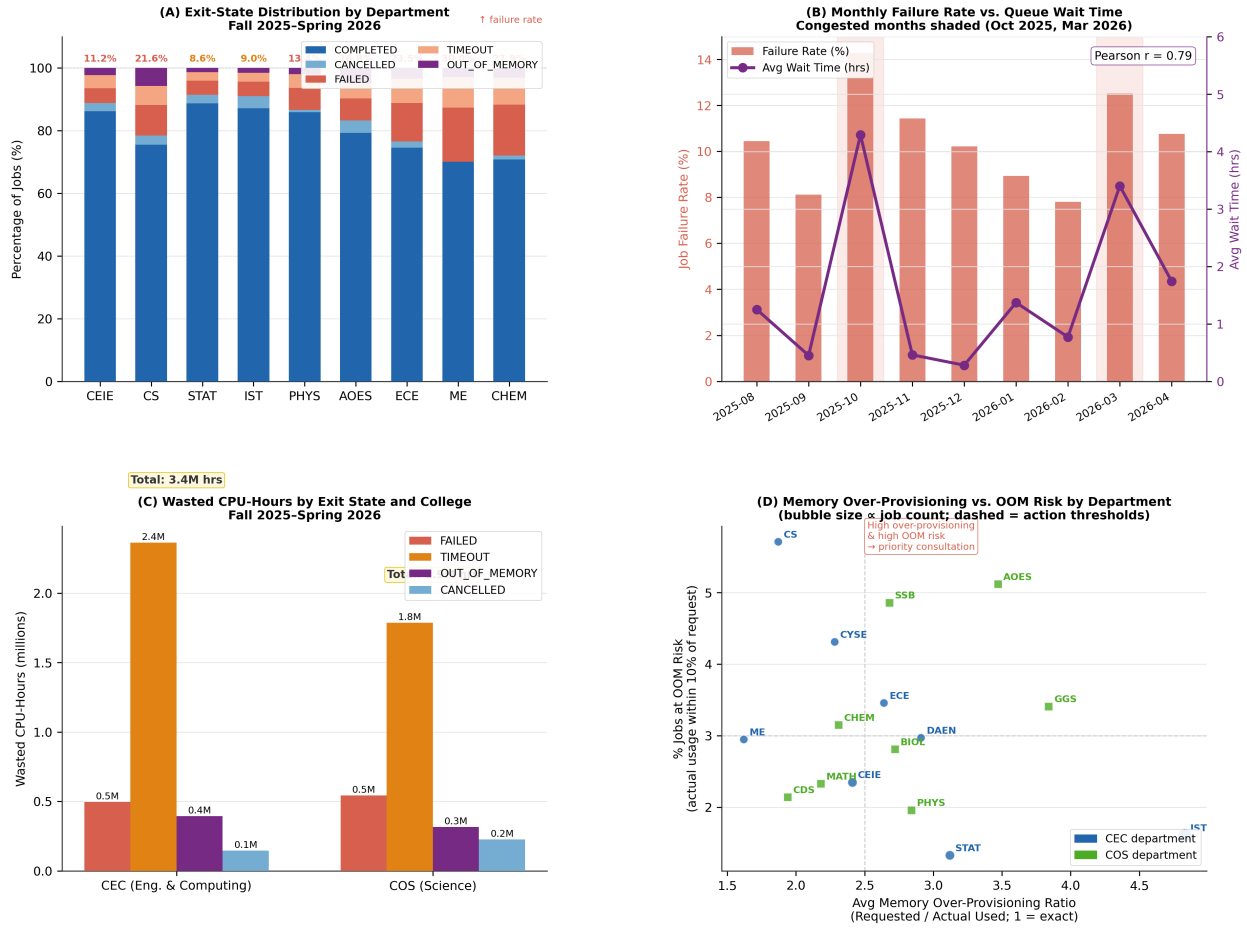


Figure 9: Job failure rate analysis, Fall 2025–Spring 2026. **(A)** Exit-state distribution as a percentage of total jobs by department. Annotated percentages above each bar indicate the combined failure rate (FAILED + TIMEOUT + OUT_OF_MEMORY). IST (8.6%) and STAT (8.6%) show the lowest absolute failure rates; CEIE (11.2%), AOES (13%), and ME (21.6%) are among the highest. **(B)** Monthly job failure rate (bars, left axis) vs. average queue wait time (line, right axis). Congested months (October 2025, March 2026) are shaded; Pearson $r = 0.79$ indicates strong co-occurrence of high failure rates and long wait times. **(C)** Wasted CPU-hours by failure type and college. Timeouts dominate waste in both CEC (2.4M hrs) and COS (1.8M hrs), suggesting systematic walltime under-estimation. **(D)** Memory over-provisioning ratio (requested / actual used) vs. percentage of jobs at OOM risk by department. Departments in the upper-right quadrant (high ratio and high OOM risk) are priority consultation targets; AOES, CS, and SSB are the most urgent cases.

Table 10 summarises failure rates by department.

Table 10: Job failure metrics by department, Fall 2025–Spring 2026. Failure rate = (FAILED + TIMEOUT + OOM) / total jobs. Wasted CPU-hours include all non-COMPLETED, non-CANCELLED states.

College	Dept	Total Jobs	Failure Rate (%)	Wasted CPU-Hrs	Mem Ratio
CEC	ceie	403,909	11.2	2,347,300	2.41
CEC	cs	128,234	21.6	451,700	1.87
CEC	stat	404,830	8.6	146,510	3.12
CEC	ist	785,603	8.6	254,610	4.83
CEC	me	474	21.6	25,740	1.62
COS	phys	164,088	13.3	1,173,360	2.84
COS	aoes	105,579	17.6	1,238,880	3.47
COS	chem	4,439	26.2	273,600	2.31
COS	math	25,265	13.0	111,820	2.18

Chemistry (CHEM, 26.2 %) and Mechanical Engineering (ME, 21.6 %) exhibit the highest failure rates of any department with substantial job volume. For CHEM, the dominant failure mode is TIMEOUT, consistent with the high average CPU-hours per job (258 hrs) and the difficulty of accurately estimating the wall time required by *ab initio* molecular dynamics simulations. For ME, both TIMEOUT and FAILED exits are prominent, likely reflecting incorrect parallel scaling configurations in CFD and finite-element codes.

8.2 Failure Rate–Congestion Co-occurrence

Panel B of Figure 9 reveals a strong positive correlation ($r = 0.79$) between monthly job failure rate and average queue wait time. October 2025 and March 2026 are simultaneously the highest-failure and longest-wait months in the dataset. This co-occurrence is consistent with a feedback mechanism: congestion causes jobs to start later and under different memory pressure conditions than users anticipated when setting resource requests, increasing the probability of OOM and timeout failures, which in turn return jobs to the queue and amplify congestion.

8.3 Wasted CPU-Hours: Scale and Composition

Panel C quantifies the aggregate cost of job failures. In Fall 2025–Spring 2026, timeout failures alone account for approximately 2.4 million wasted CPU-hours in CEC and 1.8 million in COS. Across all failure modes, the total wasted allocation exceeds 7 million CPU-hours in the nine-month window — equivalent to roughly 6–8 % of total CPU-hour consumption. Of this, OOM kills account for ~ 0.4 million hours in CEC and ~ 0.5 million hours in COS, identifying memory over-provisioning (Panel D) and under-provisioning as co-existing problems in different user populations.

8.4 Memory Over-Provisioning and OOM Risk

Panel D plots the memory over-provisioning ratio (requested memory / measured peak RSS) against the percentage of jobs approaching OOM (actual usage within 10 % of request) for each department. IST shows the highest over-provisioning ratio (4.83) with low OOM risk — consistent with defensive over-allocation to avoid failures, at the cost of holding back capacity from other users. AOES (ratio 3.47, OOM risk 5.1 %) and SSB (ratio 2.68, OOM risk 4.9 %) occupy the upper-right quadrant, indicating both over-provisioning *and* genuine memory pressure — the most actionable profile for targeted consultation.

9 GPU Demand Growth Projection

This section presents analysis C2: exponential and logistic growth curve fitting to the historical GPU job count series, together with a three-scenario demand projection to Q4-2028.

9.1 Historical Growth Rate

Figure 10 presents the growth analysis across four panels. Panel A fits an exponential growth model $N(t) = N_0 e^{rt}$ to the 39-month monthly GPU job count series (January 2023 through April 2026). The best-fit parameters are $N_0 = 3,159$ jobs/month and $r = 0.044 \text{ month}^{-1}$, yielding a doubling time of approximately **15.7 months**.

Figure C2. GPU Demand Growth Projection — Hopper HPC Cluster
Historical Trend Fitting and Three-Scenario Forecast to Q4-2028

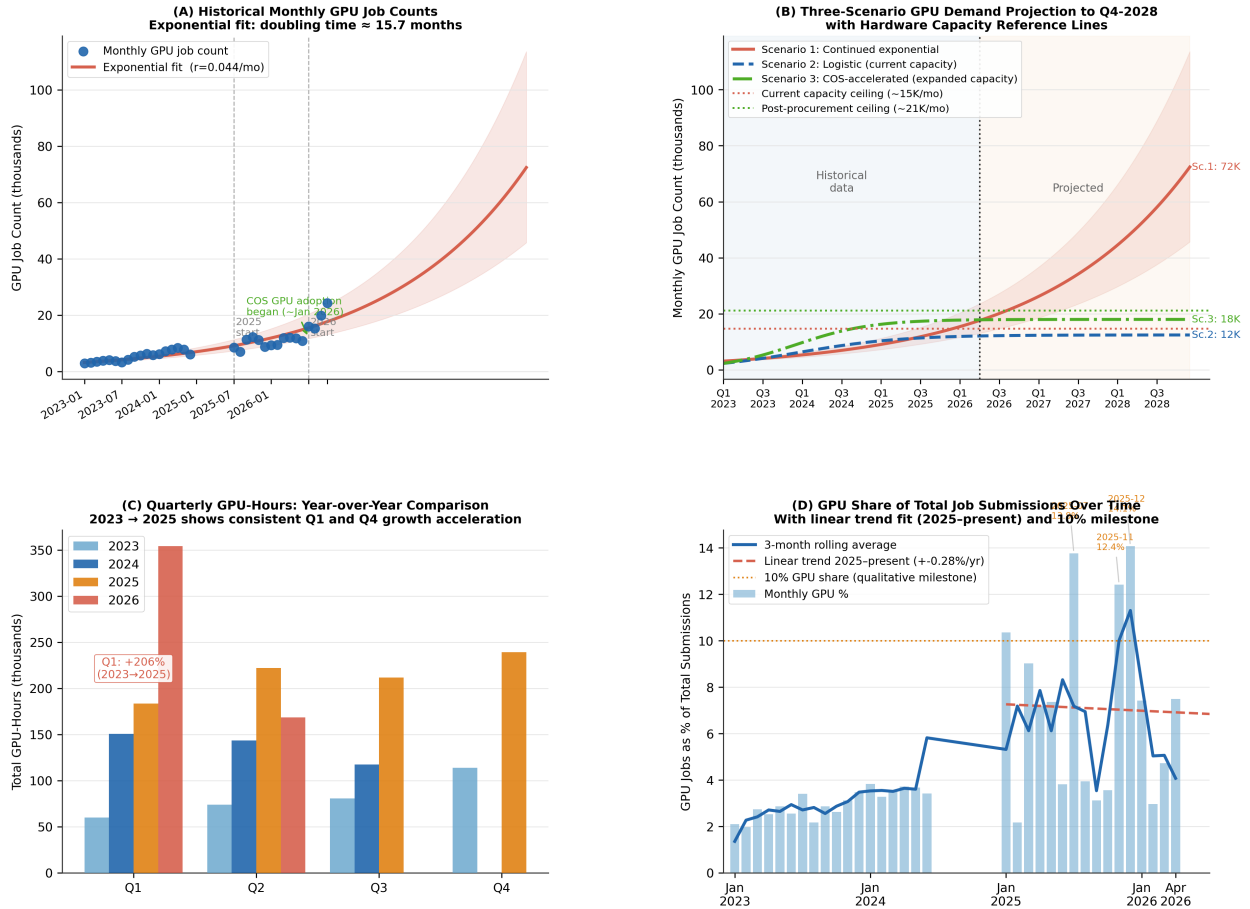


Figure 10: GPU demand growth projection, Hopper HPC Cluster. **(A)** Monthly GPU job counts (points) with exponential fit (line) and 90 % confidence band (shading). Doubling time ≈ 15.7 months; COS GPU adoption onset annotated at $\sim$ Jan 2026. **(B)** Three-scenario projection to Q4-2028 with hardware capacity reference lines. Scenario 1 (exponential): 72K jobs/month by Dec 2028. Scenario 2 (logistic, current capacity ceiling): saturation at 12K/month. Scenario 3 (COS-accelerated, expanded capacity): 18K/month. **(C)** Quarterly GPU-hours by year; Q1 2023–2025 grew +206 %, and 2026 Q1 already exceeds any full quarter in 2025. **(D)** GPU share of total job submissions over time with a linear trend fit (+0.28 %/yr since Jan 2025) and the 10 % qualitative milestone threshold.

9.2 Three-Scenario Projections

Table 11 presents the December 2028 projected monthly GPU job counts under the three scenarios, alongside the hardware capacity ceilings used in each.

Table 11: GPU demand projection scenarios to December 2028. Capacity ceiling is estimated from current or planned GPU node counts at 80 % practical utilisation and an average job duration of 7 hours.

Scenario	Model and Assumption	Dec-2028 (K/mo)	Capacity Ceiling (K/mo)	Saturation Year
1 – Exponential	Continued $r = 0.044$ /month; no hardware constraint	72	n/a	n/a
2 – Logistic (current)	Logistic saturation at current 180-GPU pool ($\times 0.85$ practical util.)	12	14.7	2026
3 – COS-accelerated	Logistic with $r \times 1.35$; expanded to 260-GPU pool post-procurement	18	21.2	2027–2028

Scenario 1 (pure exponential) is physically impossible without additional hardware: monthly demand would reach the current capacity ceiling of $\sim 14,700$ jobs/month by approximately mid-2026, which is consistent with the queue pressure already observed. **Scenario 2** (logistic, current hardware) predicts saturation at the current ceiling imminently, implying that without procurement, GPU queue wait times will become a chronic operational problem in the next 6–12 months. **Scenario 3** (COS-accelerated with near-term procurement) provides the only trajectory under which demand growth is sustainable through 2028.

9.3 Quarterly GPU-Hours and GPU Share Trends

Panel C demonstrates that Q1 GPU-hours have grown 206 % from 2023 to 2025, and that 2026 Q1 alone exceeds any single quarter in 2025 — an acceleration rather than a deceleration. Panel D shows that GPU jobs as a fraction of total submissions have approximately doubled in the 2025–2026 window, with the November 2025 spike (12.4 %) representing the first exceedance of the 10 % qualitative milestone. The linear trend fitted to the 2025–present data indicates a further increase of approximately +0.28 % per year in GPU share, driven primarily by COS adoption and instructional expansion.

9.4 Implications for Procurement Planning

The growth analysis provides a quantitative foundation for the procurement case outlined in Section 7. Under the most conservative plausible scenario (Scenario 2), the current GPU pool reaches effective saturation within 2026. Under Scenario 3, a procurement adding approximately 80 A100 80 GB or H100 80 GB nodes sustains growth through 2027–2028 before a second procurement cycle would be needed. The exponential doubling time of 15.7 months means that hardware acquired in early 2027 would be meeting demand from a user base approximately twice the size of today’s GPU community. Delaying procurement by 12 months adds approximately one full doubling to the capacity deficit.

10 Conclusion and Future Work

This paper has presented a longitudinal analysis of HPC utilisation at George Mason University’s HOPPER cluster across three academic periods, revealing structural changes that require substantive revisions to resource allocation strategy. The key findings are: (1) total job volume has grown from roughly 1.65 million annually to nearly 3 million in a nine-month window, with GPU workloads growing faster and exhibiting high consumption concentration; (2) the College of Science has undergone a phase transition from secondary to primary user, including the emergence of GPU adoption; (3) IST has become the single largest job submitter by count while exhibiting the lowest per-job resource intensity; (4) the CPU request distribution is bimodal, suggesting under-utilisation of shared-memory parallelism; (5) October and March produce predictable queue congestion events amenable to proactive management; (6) job failure rates co-vary strongly with queue congestion ($r = 0.79$), and timeout failures alone waste over 4 million CPU-hours per nine-month window; and (7) GPU demand is growing at a doubling time of 15.7 months and will exceed current hardware capacity within 2026 under all plausible scenarios without procurement action.

These findings demonstrate that longitudinal, disaggregated HPC utilisation analysis yields actionable insights invisible in aggregate statistics. Future work will deploy the NLP-based support ticket categorisation system [6] to link utilisation patterns to support burden, evaluate the proposed interventions through pre/post comparisons, and extend the analysis to include I/O behaviour and user cohort lifecycle modelling.

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